



2017 Examiners' Report

Theory Papers

This report highlights the key observations of the Examiners to provide information and help for those who sat the 2017 Masters of Wine Examination as well as guidance for future candidates. The videos of the Exam feedback interviews are published on the Institute's website and candidates are encouraged to watch the recording. It is a valuable addition to this report. Each section has been written by the Panel Chair of the respective panel of Examiners, and draws on the detailed comments made by the individual Examiners assessing the papers. Without prejudice, this report is delivered for information purposes only, and may not be relied upon by individual candidates with reference to their particular Examination papers.

Theory Panel Chair Report

Neil Tully MW

This year's theory exam produced 35 full theory passes, representing a 32.7% pass rate (2016 = 28.9%) and one single paper re-sit.

There were a good number of very strong, and indeed, a number of exceptional theory papers which were a joy to read. Well-prepared candidates writing with authority and conviction, at all times keeping their responses closely tailored to the question, achieved some excellent marks.

In contrast to the uneven performance of many candidates in last year's theory exam, this year the majority of candidates performed much more evenly across the five theory papers. This is an encouraging sign. There were also fewer cases of candidates running out of time on their last question. We also saw significantly fewer "All I know about" type essay responses, which is welcome.

Specific detail for each of the theory papers is contained within the Paper Chairs' examiners' reports, but a number of common themes have emerged which warrant comment here.

Less successful responses often lost focus. Precision of expression reflects precision of thought. Never lose sight of the question, and do pay careful attention to the way the question is worded. A few otherwise strong responses lost marks where the question was either misinterpreted, or only partially answered. The words used in the question are very carefully considered, nothing is accidental, and often indicate the anticipated structure of the response.

Do be sure to use technical terms accurately. Misused terminology undermines the credibility of your response.

Many candidates would have benefitted from reading through their finished responses and checking them before finishing the exam. Time should be allowed to do this; many errors and omissions which might have resulted from "exam pressure" could have been corrected, extra marks gained, and importantly, credibility in the eyes of the examiners would have been restored. An essay peppered with careless typing errors does not reflect the expected level of professionalism.

The examiners would like to point out that the use of the "smiley face", though entertaining, is not considered appropriate in an MW exam essay ;-(

Theory Paper 1 report

Joel Butler MW

General Comments

The range of questions for this year's Paper 1 covered the core syllabus and a number of issues and interests that the examiners believe are topical. Three questions of the six presented clearly required consideration of climate/climate change to gain a passing mark. Half of the questions posed (Q1,Q3,Q6) concerned globally important issues, requiring careful reading of the question, analysis with global examples, and an awareness of the need to focus on the main thrust of the question. Examiners observed that overall, a solid majority of candidates showed solid knowledge about a given question, but noted that there were still papers which a) failed to define well the parameters of their discussion based upon the terms of the question; b) failed to answer both parts of a question; c) made inaccurate factual statements; d) answered questions too generally and with inadequate, generic examples.

Although the average marks for all questions was a solid "C", the overall pass rate this year was less than in 2016. The Examiners suggest three areas for concern. The first is lack of knowledge about current viticultural procedures, timing and key concerns, globally. The second is not reading questions carefully and writing descriptive rather than analytical answers, effectively answering a different question(s) than asked. Thirdly, Examiners noted a number of papers that were short and lacking content, indicating possible time management issues.

1. Temperature is one of the most impactful environmental variables on wine grape growing. How does temperature affect viticulture?

A successful response to this question required addressing the statement in the first sentence as well as the question in the second sentence. The first sentence certainly required some definition or explanation of the phrase "most impactful". Good papers summed up temperature's impact on the vineyard nicely in one paragraph. But the actual question in the second sentence seemed misunderstood by several candidates.

Indeed, as one examiner noted about one answer: "...more a discourse on how climate affects vineyards at different points, with too few relevant connections to actual responses in the vineyard to temperature/climate issues." This question sought the ability to connect temperature variability to specific practices in the vineyard, relative to the type of wine or grape varieties produced.

This question was not about climate change per se (though good answers addressed that issue in a coherent manner). Examiners noted instances of what seemed like a stock response on climate change, without any attempt to tie it in to the question. Passing papers correctly made the connection between practical responses in the vineyard during the year to temperature-driven situations in different wine growing climate zones.

Some papers revealed a lack of 'real-world' understanding of how to respond viticulturally to the seasonal variations in a vineyard brought on by temperature, engaging examples from books rather than from concrete examples reflecting direct experience or communication with producers around the world. Simple factual errors are now less common, examiners noted. However, in some cases confidence was undermined by ignorance of, for example, the lack of knowledge of relevant temperatures and timings which might cause plant growth to stop, the forthcoming crop to be lost or the death of the whole plant.

Examiners were pleased to see (in the words of one Examiner) "nice global knowledge, and ties to the question, and ...understands the key issues." Important issues that could have been considered and discussed (with the necessary viticultural responses) for a passing paper included: temperature variability, individual varietal tolerance to low and high temperatures, temperatures needed for ripening a balanced crop, air and soil temperatures, warmer and colder soil structures and their effects, the influence of humidity and possible climate change effects on individual grape varieties.

2. Water availability is increasingly an issue in some wine producing regions. How can a viticulturist best ensure water sustainability when establishing and managing vineyards in drought-prone regions?

This question was one of the more successfully answered questions on Paper 1 this year. Less successful papers failed to provide definitions where required, did not answer both parts of the question, or failed to understand the question's intent. Differentiating between 'availability' and 'sustainability' carefully distinguished the better papers. Good papers resisted the temptation to describe "all I know about" water's importance to growing grapes. The critical definition of 'drought-prone' was sometimes poorly addressed, leading to discussions about regions where water sustainability was a factor more due to short term fluctuations in weather, rather than long-term conditions yielding perennial drought. The best papers clearly distinguished and explained the differences between water needs for establishing a new vineyard as well as that of an established vineyard.

Good papers managed to cover key areas such as the potential need to change rootstocks or scions, reform vineyard canopy management/training regimes, remove water-based frost protection systems, change timing/type of fertilising methods, and deal with legal restrictions. A strong response required examples from around the globe, not just from one or two areas well-known to the candidate.

3. Labour supply for vineyard work is decreasing in many parts of the world. If this trend continues, how will this affect viticulture, and how can vineyard managers around the world best prepare for, and handle, a shortage of workers?

This question had the greatest number of responses on Paper 1. The inclination might have been to answer it as a Paper 4 (business of wine) question. Indeed this happened frequently and accordingly the pass rate for this question was the lowest on the Paper.

The question required global consideration of labour availability relative to the needs of producers to maintain profitability. Conditions might range from more restrictive US immigration of Latin American farmhands to the inability to rely on workers in war-torn areas owing to their migration to nearby wine-growing areas such as Lebanon. Above all, a good paper had to consider how mechanisation can provide solutions across a spectrum of viticultural needs to alleviate a shrinking labour supply. A strong response may have included how to better manage labour year-round, for example by providing education and training to perform a range of vineyard tasks by a smaller group of permanent workers. Passing papers overall reflected what one Examiner noted: “Good examples and demonstrating knowledge in key areas of mechanisation plus sufficient discussion on training and multi-discipline approach to work.”

Summing up, a passing answer acknowledged and discussed the various viticultural tasks (across the life of the vineyard from establishment to grubbing up, as well as during the grape/wine growing year) that can be automated or mechanised, as well as those that cannot.

4. Discuss which vine varieties would be most suitable for the production of dessert wines. Explain your choice with specific reference to any climatic requirements.

This question called for a good demonstration of candidates’ global experience and knowledge. It required a range of viticulture practices to be considered within a framework of options that address the climate requirements for the production of dessert wine styles.

To earn a passing mark, it was vital not to ignore fortified wine. It was also critical to understand “dessert” wine as a wine with the assumption of sweetness, and an awareness of the levels of sweetness across a wide range of varieties and wine styles. Finally, remembering this was a Paper 1 question, good answers clearly linked variety and style of wine to its climate requirements. This included choice of variety based upon criteria such as skin thickness, rate of sugar accumulation, acidity and aromatic compound levels, production method, and humidity levels. The best papers noted that the relationship between climatic requirements and varieties flowed in both directions; that is, varieties adapting to their climate niche over time (Semillon in Sauternes) or climate requirements dictating adaptation of a particular grape variety based upon the unique production methods necessary (PX in Montilla-Moriles).

Most papers answered the first part of the question well enough, but the link to the climatic requirements sometimes broke down. There was overall, good discussion of various varieties and why they made for good dessert wines. Sometimes, however, that discussion didn’t reflect enough consideration of the many viticultural reasons why they were ideal dessert wine grapes. Examiners noted that several papers merely ‘surveyed’ the varietal scene for dessert wines, but made no

attempt to answer the question. The huge global choice of wines/varieties to discuss necessitated making choice about 'the most suitable' varieties to discuss as the question demanded. For example, several papers were convincing regarding varieties, botrytis influence and the connected climatic issues, but some erred, citing Furmint, Semillon, Riesling and Chenin Blanc as all successful due to their thin skins.

Top marks went to those papers that strongly linked a variety's characteristics to the climatic considerations that made it the right choice for that place, and for the method of production best suited to expressing character. One Examiner noted, "As we know, dessert wines can come from many diverse climates, and the weather conditions from vintage to vintage often stray beyond the 'norm'. But to try to grapple with both those variables at once in an essay this short was very tough." Therefore the examiners tried to be flexible and generous in marking, so long as factual errors were absent, and most of the categories (still, fortified etc) were considered in the discussion with relevant examples.

5. Does soil preparation affect the potential yield and quality in a vineyard?

Question 5 prompted the fewest responses (34) on Paper 1, perhaps reflecting a lack of knowledge by candidates of this important viticultural activity, or perhaps because many felt intimidated by the rhetorical nature of the question.

The Examiners noted this question as one requiring sound knowledge of viticulture to answer well. The question asked the writer to address the role played by the soil when it comes to providing support for the vine throughout its life cycle, its ability to provide the vine with water, and physical support for the vine. Candidates were asked to consider the various options available to vineyard managers when it comes to soil preparation, and importantly the impact of these interventions on yield and quality. Several papers answered a different question in their authors' minds, along the lines of "Describe everything you know about why soil is important". Issues such as annual pre-season preparation or biodynamic preparations also fell within the scope of this question.

It was critical to understand that Soil Preparation encompasses major physical steps such as ripping, weed control and soil pH adjustment, as well as major "chemical" steps such as addition of nutrients such as Potassium or Nitrogen; about why, how and when these should be done and the effect on quality and yield. Equally important was the issue of water availability and the physical support requirements of the soil for vines and then referencing the different means of preparation relative to the type of wine being produced, be it Château Latour or Pinot Grigio del Veneto. Global examples were critical to obtaining a good pass.

Successful papers included discussion of established vineyards, as well as commentary about new technologies now used to help with soil preparation. A number of otherwise solid papers covered physical preparation in detail, but omitted nutrient and water issues. Examiners did note that with a question like this one, not identifying "experts" did weaken an argument.

6. What are the principal pests and diseases facing vine growers today, and how can they best be managed?

A popular question and the most successfully answered overall. Critical to answering it well was defining 'principal', 'pests and diseases', and 'manage'. If one didn't do some form of triage and focus on those that are 'principal' around the world today, and knowing what to do about them, it was very easy to "not see the wood for the trees".

Examiners are aware of the sheer number of issues affecting vine-growers globally, and thus set a broad range of pests and diseases that could be considered, giving candidates the opportunity to focus on 7-8 of which they considered important. This was a question in which the best papers clearly highlighted the writer's real experience due to direct knowledge from producers, and demonstrated their global awareness of the situation.

Good passing papers considered the differences between physical pests such as deer, and those pests transmitting diseases: nematodes, leafhoppers, mites, mealybugs. Likewise, it was critical to discuss (and distinguish) between key bacterial/fungal diseases such as Pierce's Disease, Flavescence Dorée, botrytis, Eutypa, Esca, and viral ones such as fan leaf virus, red blotch and leafroll. The key to a pass here was clearly presenting arguments and examples demonstrating an understanding of managing and controlling the said disease. In the words of one Examiner commenting on a particularly good paper: "A very detailed answer, well organised and peppered with relevant examples. It defined 'principle' or how the pests/diseases discussed were selected, considers costs and considered biodynamic/organic treatments as well. It mentioned most of the pests and diseases, and detailed how they could be curtailed." Many papers (not all!) nominated phylloxera as (still) a principal pest. So long as the discussion of management was relatively short and pointed out that it is not a significant threat in most places due to better knowledge, examiners were positive in their marking consideration.

Discussions of management needed to consider varying growing conditions and the philosophy of the producer; the best papers confidently discussed management options open to organic/biodynamic growers as well as the less restrictive ones available to more conventional growers.

Theory Paper 2 Report

Meg Brodtmann MW

General Comments

The Paper 2 Examiners made the same general comments that they have made every year and these are bullet pointed below to ensure they are clear.

- Paper 2 is a technical paper and you cannot pass without including technical details!
- Spelling. If a candidate cannot spell wine regions, winemaking terms etc., it does not imbue much confidence in the examiner. One of the key aspects of the MW is communication and you cannot communicate with poor spelling.
- Answer the question! Do not brain-dump to fill the paper with words.
- Listing examples that are out of date indicate that the student is not aware of the current state of the industry. Examples must be relevant, global, varied and support the argument rather than be used as the argument.
- If you are not sure of the facts, don't put them in.
- Do not just focus on the premium end of production, students should also understand how large volume wines are made.
- Too many students use the technique of saying "who does what where" but do not show any critical thinking. This is linked to brain dumping but we are invariably asking you to take a position and critically apply your knowledge.
- Handwriting needs to be legible.
- Do not use derogatory terms or make harsh judgement calls on wine styles, regions, etc – the winemaker may be marking your paper!

1. 'This wine has been bottled unfined and unfiltered'. Assess the implications for the winemaker.

A question that tested candidates' knowledge of what procedures are available, what is legal in different parts of the world, what the difference is between fining and filtration (which wasn't always clearly addressed!) and – crucially – what other winery activities can be undertaken, or are implied, if fining and filtration are forsaken. The major over-riding omission was the simple failure to talk about how things like yields, sulphur dioxide regime and vat/barrel ageing might be deployed to achieve greater stability and clarity in the absence of fining and filtering. The more successful responses to this question looked at the whole range of winemaking procedures that can be employed to eliminate the need for fining and filtration. Confusion over the difference between fining and filtration was also quite surprising – candidates must have clear definitions of these two processes before starting a question such as this.

Quite a few candidates preferred to consider the labelling and marketing implications of what might be considered bad winemaking rather than the technicalities of winemaking, which is what this paper is about. It also seemed difficult for students to discuss the implications for the winemaker of making unfinned and unfiltered wine. A list of who is doing what, as opposed to a technical approach, is not enough.

2. Sauvignon Blanc appears in many styles throughout the world. Compare and contrast Sauvignon Blanc winemaking practices around the world.

Some excellent responses to this question showed good knowledge and understanding and got to the essence of why wines are different from the same variety and how, with a variety such as Sauvignon Blanc, a winemaker can embrace a great many options if they have the knowledge, skill and the courage to step away from a mainstream trend.

Candidates should however beware of overgeneralisation. The examiners commented on the very frequent reference to only a few examples, for example Fumé Blanc in the USA without considering newer producers who are making complex Sauvignon Blanc in that country. Examiners were looking for a discussion of reductive versus oxidative winemaking, use of expressive yeasts, use of oak, international examples and blending. The candidates needed to back up their comments with sound technical information. A small number of essays gave welcome thought to technical winemaking such as separating press fractions, types of fining practices, how residual sugar may be employed, and stabilisation. Some less well prepared candidates tended to waffle and focus almost solely on high quality, terroir-driven wines while neglecting the presence of high volume wines.

This question gave the candidates the opportunity to demonstrate their reasoning with use of international examples to support their arguments. Few took this opportunity. It also gave the candidates a great opportunity to cover off a lot of winemaking practices in one essay.

3. Discuss the role of enzymes in winemaking.

This was one of the less popular questions on this year's theory paper two. However, a competent candidate had the ability to really shine here and get high marks whereas those candidates that were less confident on the subject found this question extremely challenging. This was reflected in the marks achieved – good passes or clear fails with very little in between.

Those candidates that did not do well failed to define what an enzyme is and what it does. Like so many other essays, candidates should have started at the beginning of the winemaking process and worked their way through discussing the use of enzymes at the various steps and stages. Candidates needed to discuss red, white and rosé winemaking. It was interesting to note how few candidates mentioned enzymes and red winemaking, demonstrating a big, general gap in knowledge.

Commercial enzyme types were mostly covered but few, if any, candidates discussed spoilage enzymes present in grapes. Again, a systematic approach covering the whole winemaking process would allow candidates a structure on which to hang their points.

4. Compare and contrast winemaking techniques best employed for Syrah grown in Cornas, McLaren Vale and Hawke's Bay.

Although few candidates attempted this question, many papers were competent with few really poor answers. This question required a good understanding of winemaking in the regions described. Clearly the students mostly understood the winemaking techniques of the Rhone/Cornas and some were conversant with McLaren Vale/or warm climate areas, but fewer were conversant with Hawke's Bay.

At first this may have felt like a red wine version of Question 2. In fact, it opened up into a far greater invitation to the student to accurately describe what makes McLaren Vale distinct from other classic Australian regions such as Barossa, and what makes Cornas distinct from other classic Rhone appellations. This was a huge micro-terroir question, potentially allowing students to compare and contrast the different winemaking techniques in the three regions, and how they impact on quality and style. Very few answers addressed this, with most electing to describe McLaren Vale as if it were part of a gigantic generic Australian whole and Cornas as if it were entirely representative of all Northern Rhone reds.

5. Temperature management is the key to modern winemaking. Discuss.

The main issue with this widely-answered question was that many candidates tended to focus on the temperature management of fermentation only and thus lost marks by not elucidating other areas where temperature management is important. On a question such as this, temperature values do need to be included. Either these were missed out or candidates hedged their bets with larger ranges: 10-16°C.

Overall this was a wide-ranging question that needed to touch on many different aspects of winemaking – from fruit storage, processing, fermentation, through to long term storage. Very few candidates covered all these areas. Examiners commented on lack of sound technical understanding of must chilling. Cold soaking on white and red wines is not an equivalent term.

Better answers always employ the chronology of the winemaking process. Considering the entire winemaking process in a systematic manner and thinking about the role of temperature management at each step will ensure that no points are missed.

6. How, and to what extent, can yeast influence the final style and quality of wines?

This question was not asking for “Everything I know about yeast”. We urge candidates to address the question and not just tell us everything they know about a particular topic.

Examiners also remarked on some candidates’ ability to write an entire essay on yeast without once mentioning a single species. Paper 2 is a technical paper and not only do candidates need to know the names of common yeast types involved in the winemaking process (be they good or bad!) but they also need to be able to spell the names of the yeast correctly.

Narrow minded comments such as “selected yeasts only produce short lived wines”... or “Spoilage yeasts and Brett can be used to bring out terroir”... do not suggest a deep understanding. Nor do generalisations about old world and new world wine styles.

Questions related to yeasts appear with some regularity so it is surprising to see candidates still failing to grasp the role of the different yeast genus and species. This is one area where much R&D is spent so students would be wise to keep themselves up to date with what is going on. It is also wise to ensure the examples used are global and relevant.

Theory Paper 3 Report

Caroline Gilby MW

General comments

Paper 3 had a relatively high pass rate of close to 40% in 2017, so though it may not cover the most glamorous topics in wine, many candidates who have appropriate technical knowledge do well. There are relatively few very high grades but these show clearly that strong answers are possible.

On occasions candidates failed to answer at least part of the question as set and were tempted into simply writing what they knew about a topic. Lack of numbers and real world practical detail or examples was another common criticism. These of course must be used to support reasoning. Time wasted with overlong introductions was a problem noted last year but this does not seem to have been an issue in the 2017 exam. However, it is worth repeating that a good introduction should summarise understanding of the question and the ground to be covered, as this will help candidates focus.

Paper 3 is a technical paper and it is expected that candidates will have a firm grasp of some key factual and numerical information. EU limits for SO₂ use, typical levels of free SO₂, actual values for pH, permitted and actual usage levels for other additives, pore sizes for filtration and so on are key knowledge for MWs. Vague comments like “lower levels of SO₂” are not sufficient – actual numbers are required. Inaccurate use of “Sulphur” instead of sulphites or sulphur dioxide still crops up regularly, and while this may be widespread shorthand among winemakers, there are important differences between sulphur as an element and its compounds.

1. Outline the most important quality control checks during bottling and why each is necessary.

Key understanding:

Recognition of:

- *Most important risks inherent in bottling process, how and when these arise*
- *Consequent need for and timing of quality control checks*
- *What these checks comprise*
- *Relevant regulation and voluntary codes relevant to quality control*
- *Commercial consequences of failing to address these risks adequately (implicit in the wider scope of “why each [check] is necessary”)*

Candidates should be aware that bottling (to include alternative packaging) is the final step of wine production during which the wine can be microbiologically compromised or damaged by oxygen ingress. Bottling isn't solely about putting a wine into bottle, it is also about ensuring the correct wine/variety is bottled, correct bottle type is used and the correct packaging is applied.

There were few really strong answers to this question and lots of candidates lacked focus or missed too many key issues. Candidates needed to consider both quality checks on wine and checks on the packaging process and dry goods, as well as compliance with product specification. Too many candidates simply described the bottling process and failed to consider the quality control part of the question, showing lack of awareness of key QC analyses for wine and why they are carried out – essential knowledge for potential MWs. Consideration of HACCP is always useful in a question that mentions bottling. Quality control is part of the overall quality assurance process and wine is a food stuff which must comply with legislative and regulatory standards (so an awareness of EU legislation and ISO requirements is helpful). A lot of candidates missed the critical “why” part of the question, and without considering this, their answers were too superficial and lacked analytical depth. The safety of staff handling the liquid and the packaging is critical (risks like broken glass, noise, heavy lifting and exposure to health risks). Issues for the wine itself include loss of quality, consumer disappointment, damage to brand, and costs attached to issuing recalls or being unable to sell the wine.

2. What analyses should be carried out at the winery pre- and post-bottling to ensure that a wine is in good condition and conforms to legal requirements for sale?

Key understanding:

Key elements of the answer should show understanding of what can theoretically go wrong at the bottling stage, and how analysis can help prevent wines being bottled in poor condition and within the legal parameters. An understanding of the methods that are widely used and the legal limits for the various constituents of wine is required.

This question had the highest pass rate at 50%, indicating that it was straightforward for candidates with good levels of technical knowledge. This question required knowledge of key quality control analyses establishing a wine to be in good condition, both before bottling and after. It also shows whether it is likely to remain so after the bottling process and until it reaches the consumer. Specifically, the question asked “what analyses” yet there were disappointingly few answers which listed and explained the key QC analyses and their importance in ensuring the wine’s good condition and conformation to legal requirements. Several candidates simply discussed issues that might arise and not what QC checks were required. Here again the key problem was the overall lack of technical information and analytical reasoning in many of the answers. The comment “too qualitative” was used by examiners for many of the scripts.

Candidates also needed to know legal limits (EU or other key markets) for key wine constituents and additives, plus allergens. Knowledge of labelling requirements was also required (e.g. sweetness declarations such as medium-dry or halbtrocken, allergens, sulphite declarations, alcohol tolerances). Supporting examples would be helpful too.

3. Many winemakers are reducing the levels of free and total sulphites in wine. Consider the role of sulphites at bottling and until the wine reaches the end consumer. What are the implications of reduced levels of free sulphites?

Key understanding:

- *Risks arising at bottling and continuing until wine reaches consumer.*
- *The four distinct properties of SO₂ - anti-oxidant, anti-microbial, anti-oxidasic and (less important) binding with acetaldehyde.*
- *Focus on the role of SO₂ at bottling to the end consumer and the implications of lowering the level of free sulphur dioxide – i.e. wine not being in drinkable condition or developing prematurely, or developing faults such as microbial fermentations.*
- *The role of free and total sulphur dioxide - in various wines, including numbers.*
- *Potential risks to health at high levels of free SO₂, consumer concern about additives, especially since requirement for obligatory sulphite labelling. Lower levels may improve aromatic expression in young wines.*
- *Natural wines made with no added sulphur dioxide, or very little. Customers for such wines may have higher tolerance for what some would see as faults.*

The “implications” part of the question required a definition of what “reduced levels” of free SO₂ mean. A free SO₂ level below 10ppm will run the risk of the wine oxidising and be a concern to the winemaker, unless it is a natural wine that is being produced. Free sulphur dioxide has a major purpose in preventing wine from oxidising both at the time of bottling (dissolved oxygen pick up) and during the life of the wine. It also helps maintain the wine’s freshness plus inhibits yeast and bacteria, and thus reduces risk of refermentation in bottle (including malolactic fermentation) and these are essentially the implications of reducing the level of free sulphur dioxide at bottling.

Implications in managing risks: crucial importance of aseptic state of all bottling equipment, bottles and stoppers; reduced temperature of bottling environment, storage and transit to consumer; inherent energy cost. Storage and transit after leaving producer’s premises are out of producer’s control and so a particular risk.

It is still surprising how many candidates do not know current EU levels for total SO₂, nor typical levels of free SO₂ in different styles of wine, nor what constitutes lower levels in actual numbers. It is really important to have these facts committed to memory. Clarity about the difference between free and total SO₂ is also important.

It was clear that quite a number of otherwise good candidates misread the question: instead of addressing the “implications of reduced levels of free sulphites”, they wrote about the implications for the winemaker of reducing free sulphites/how to achieve low sulphite wines. Their essays thus focused on adjustments to be made in the winemaking process, so treating it substantially as a Paper 2 question. These essays would typically score in the late 50s/early 60s but would not pass. There were, however, lots of very good scripts which did address (both parts of) the question directly and met the standard to pass.

As a general comment on the answers to Q3, one examiner expressed the view that “there were many candidates who appeared either to have pre-prepared an essay on the use of SO₂ and who rolled it out with disregard for the focus of the exam question or who mis-read the question; consequently there were a number of candidates who, in the abstract, appear technically knowledgeable but who have not attained a pass mark.”

4. Write concise short notes on 3 of the following topics:

- a) **Preservatives in wine**
- b) **Preventing deposits in bottled wine**
- c) **Inert storage**
- d) **Pre-bottling filtration**

Key understanding:

Straightforward technical answer for candidates with good knowledge base.

Those candidates who failed this question usually did so because their answers were insufficiently wide and/or deep in scope in addressing the chosen topic. Writing ‘short notes’ still has to evidence sufficient knowledge and understanding for MW level and use supporting examples. Quite a number appear to have left this question until the end of the exam and may not have left themselves enough time to do justice to it. One examiner commented, “in contrast, there were some outstanding answers, which were a joy to read.”

Preservatives in wine needed to cover more than simply sulphites. Alcohol itself is significant for its preservative action and other preservatives like ascorbic acid and potassium sorbate also needed to be covered. Mention could be made of lysozyme and Velcorin for their preservative action too. The action of ascorbic acid and sorbic acid was muddled by several candidates

Deposits in bottled wine was often taken to mean tartrate crystals only. Other deposits such as protein, metal casse and microbial hazes needed to be mentioned. Inert storage answers were generally sound. Filtration was well answered by some, but anecdotal and too general by others and often failed to cover anything but pre-bottling membrane filtration.

Theory Paper 4 Report

Simon Thorpe MW

General comments

In this year's paper we looked to examine some specific areas of the wine industry, whether that related to return on investment or to commercial issues within a single region or country. The topics covered were all very mainstream, but the answers sought detailed knowledge of certain aspects of those markets. I sense that this might have caught out a lot of candidates who had prepared to a somewhat superficial level and therefore whose examples and detail were found lacking. While there were some excellent candidates, overall the level of achievement was lower than last year. It is not sufficient to play safe – it is vital to shine and stand out. One thing I'd suggest is a real focus on learning facts and figures with greater accuracy – there seemed to be far more vague or plain inaccurate numbers in the answers this year. Furthermore, as is often the case, a simply argued, well structured and concise answer will stand more chance of success than long, rambling and unstructured answers. DETAIL and ACCURACY are key factors.

1. How can the role of the intermediaries between producer and consumer be justified? How is it changing? Use examples from several different wine markets.

Key Understanding:

An accurate definition of key terms, particularly 'intermediary': (agents, distributors, wholesalers, sole traders, brokers). Perhaps a brief explanation as to why the supply chain for wine differs from other grocery/FMCG/agricultural sectors, and in different countries. Note historical/legal/economic factors.

This may well lead the candidate to take a PESTLE or SWOT approach, or alternatively to analyse the benefits/weaknesses of each element of the supply chain in turn from producer through to consumer. The title indirectly questions the role of retailers and thus leads candidates to consider 'direct to consumer' as a route to market and thus an exploration of such emerging models (this helps to explain how routes to market are changing) especially through on-line channels. (Examples might include drop-ship, market place, set up by LVMH of 'Clos 19', Chateau Margaux opening wine bar).

Overall, a good pass rate of 41%. Candidates who scored well provided well-structured answers with concise analysis from a number of markets. Those that didn't fare so well were generally long on describing the market and the intermediaries that act in it, but missed out on analysis of the causes and outcomes of the changes that are taking place. One key factor which requires attention in preparing for the exam is ensuring more than a superficial knowledge of a number of different markets and routes to market around the globe. There was a large percentage of candidates who knew one market intimately, but this was not sufficient for a pass. It was also felt by the examiners that, although there is no one "correct" answer to this question, there is a general move towards intermediaries becoming less relevant in an economic/financial sense, so the best answers made reference or explored this theory.

2. Can small independent wine retailers compete with large chains on price? How else can they compete effectively?

Key Understanding:

- *There are small independent wine retailers in all markets who carry a bigger range than the super/hypermarkets/retail chains.*
- *Independents can't compete like for like on price - but have significant other advantages.*

This was not a complicated question and it was generally well answered with more than half of those that attempted it achieving a pass mark. The best answers again had a range of examples from several markets, explored the financial aspects of managing suppliers and margin expectations, as well as looking at the non-empirical elements of running an independent retail outlet. If the basics were covered when defining the key terms in the question and laying out a structure then it proved fairly straightforward. The main weaknesses were superficiality of examples and conclusions and, as with most of the questions in this paper, a lack of understanding of more than one marketplace. This proved to be a good MW question for assessing whether candidates had a depth as well as breadth of understanding.

3. You inherit the equivalent of US\$ 10 million and want to invest in the wine industry. How would you spend your money to attract the greatest return on investment? Explain your logic.

Key Understanding:

This was a far-reaching question, which could be interpreted in many different ways. It therefore required a strong, supportive structure to avoid drifting. Return on investment (ROI) needed a clear definition and a consideration of short, medium and long term horizons for accurate assessment. The risk was that candidates drifted and wrote too much, or veered off subject and wrote about a specific opportunity without considering wider options and context. (Fine to ultimately conclude on one opportunity but some broader perspective should have been given).

Logical structures went through the elements of the supply chain from producer through to retailer across different channels and different markets. Alternatively a SWOT would have kept the candidate on track and demonstrated rigour and logic in their thought process. A brave but potentially narrow approach would have been a financially driven case.

Answers to this question achieved the lowest pass rate at below 20%. It was designed to test understanding of financial choices in the wine industry and there's no doubt that those candidates who were able to demonstrate that understanding were the successful ones. It's also worth noting that this is a Paper 4 question, not Paper 5, so those candidates who were too dreamy and imaginative, without the financial numbers to back up their argument, were the least successful. Any industry, even the wine industry, is founded on profit generation and this was a bald, bold question asking about financial return on investment.

One further point to make is the numerical inaccuracy of examples used in many answers. If this was Paper 1 or 2 then the technical aspects of wine production (maybe g/l of RS or yields per hectare) would absolutely need to be quoted accurately – the same applies to this paper for financial details.

4. In what ways has social media changed the marketing of wine brands over the past 10 years?

Key Understanding:

This is a broad question that would require as a starting point a strong definition of social media. Strong candidates will recognise that there are industry studies e.g. from 'Wine Intelligence', looking at the role of social media in wine. This could help provide an evidence-based framework for answers which may have the potential to be too subjective.

A key element of this question is to consider the overall marketing of wine brands and how social media has assisted (or not) and whether more classic marketing tools still prevail.

Perhaps not surprisingly, this was answered by many candidates, but the pass rate was fairly low - just under a third. The comments from the examiners praised candidates who were able to provide analysis and insight in a well-structured and focused answer whilst condemning those whose arguments were hard to follow, too long and rambling. The key factor, as is often the case, was being able to assess which structure best suited the subject. From there, it was important to ensure that all the relevant points are covered and backed up with empirical evidence. This was not a question which asked for a list of social media outcomes, but specifically how social media has impacted wine brand marketing over a period of time.

5. You are Export Director of an established large wine company producing in excess of one million nine litre cases. How would you seek to capitalise on the potential of the growing Chinese market for imported wine? What are the risks and how can they be managed?

Key Understanding:

- *Size of the wine market in China and current trend of increase. Segmentation by price band.*
- *Structure of the importer/distributor network - single national importers or multiple regional players.*
- *Breakdown red/white/rosé/sparkling.*
- *How do Chinese consumers access information about wine?*
- *Demographics. Male/female split, age group preferences.*
- *Consumer perception/understanding of grape varieties, prime regions, dry/sweet styles.*
- *Credit risk, Counterfeiting, Price control, Distribution channels, Language barrier.*

A fairly well answered paper – just over 40% passing. This question couldn't be answered well unless the candidate had a good knowledge of the Chinese market, the trends/developments in the market and crucially the risks associated. There was little point attempting to answer this without that depth of understanding and yet still quite a few gave it a stab. China is such a key factor in the business of wine these days that it must be studied in some detail – knowledge of the market would have helped in every single question in Paper 4 this year.

Good answers covered the bases with good examples. Great answers added in elements of the different channels/routes to market. In general, the more detailed the knowledge, the more successful the answer.

6. As the owner of a Bordeaux Classified Growth from the Left Bank, what options are available to you today to present your wine to the market? Evaluate these options from commercial and brand-building standpoints.

Key Understanding:

- The use of Bordeaux négociant network.
- En primeur.
- Withdrawal from the en primeur market, e.g. Latour, releasing stock at a much later date.
- Combination - some en primeur, some later releases, drip-fed into the market.
- Employing one's own salesforce. Analysis of benefits against additional costs.
- Appointing nominated importers in selected markets. Exclusivity or multiple importers?
- Cellar-door sales. Oenotourism.
- Financial grasp of the costs of distribution, and margins achieved by importers and négociants.

A classic question about the most established wine market. Sadly, a pass rate less than 30%, with most of the lower marks showing a lack of awareness either of the changing environment around en primeur or of the detail of the workings of the prevailing system. Also a lot of answers focused on either one or other of the commercial and brand building standpoints, but not both. The word “today” in the question implied that you should consider the contemporary issues, comparing them with the situation in the past. Again, successful questions showed deep knowledge of the mechanics of the market, the options available, with numbers-based evidence. Brand building also needed some numbers to illustrate success or failure.

Theory Paper 5 Report

Beverley Blanning MW

General comments

The general standard of essays on Paper 5 this year was good. The questions are designed to encourage wide-ranging answers, although any answer must remain within the confines of the question. Perhaps the most common failing on this paper is not adequately answering the question asked.

Essay structure remains a problem for some candidates. This is especially important in this paper because we are examining candidates' ability to construct a meaningful argument. That said, there is room for creativity and there are certainly no fixed model answers. One candidate successfully managed to address one question by means of a short story - admittedly a risky strategy, but in this case one which paid off.

It may sound like an obvious point to make, but essays containing basic incorrect facts or sloppy use of language are unlikely to pass. Any quotations used need to be correctly attributed, the examiners will pick up on such things, therefore it is recommended that students only quote when they are sure of their source. One final general word of advice to candidates is to check their scripts when they have finished. There were many more spelling errors this year, both of basic words and wine words. This appears to be caused by careless typing.

1. What is the importance of alcohol in wine?

Key understandings:

- *The central importance of alcohol to wine, technically and legally.*
- *The quality and sensory importance of alcohol to different styles of wine.*
- *The cultural importance and impact on the individual and society.*
- *Clear recognition and communication of the positive and negative aspects of alcohol in wine.*

This was probably the most straightforward question on the paper, so it was surprising that the pass rate was not higher. The examiners felt that the question was not well answered, perhaps because there was a lot of ground to cover and many candidates neglected to demonstrate their understanding of significant key areas. A broad and discursive approach was needed and candidates failing to take this approach were unlikely to achieve a pass.

We look for analysis, not description. However, there is a wealth of factual information on the subject that MW candidates need to have at their fingertips to answer this sort of question. We were expecting to see evidence of understanding alcohol's importance for microbial stability, preservation and longevity, the legal and tax implications, and the control of alcohol consumption by governments. Awareness and discussion of alcohol's contribution to structure, balance and style, and the importance of its interaction with other elements in wine were also essential components of

a good essay. A number of candidates focused almost exclusively on the health aspects of alcohol in wine, which could certainly form part of a good answer, but would have been insufficient to ensure a pass. There were many bland and over-general answers that showed neither critical thinking nor the breadth of knowledge we were looking for.

2. Has science taken away the romance of wine?

Key understandings:

- *The question concerns the juxtaposition of science and art: should one or other be dominant? Can they co-exist?*
- *Understanding of romantic notions of wine, past and present.*
- *The role of scientific developments in changing perceptions about wine.*

The question required candidates to consider both science and romance as they relate to wine. It was imperative to consider both sides of the question, rather than present a one-sided argument. The latter approach was taken by quite a number of unsuccessful candidates.

Romance may refer to any number of intangibles, such as the mystical origins of wine, current notions of glamour and exoticism, attitudes to artisanal production, unpredictability of harvests, and so on. Candidates needed to consider for whom science may or may not have taken away the romance of wine: producers, consumers, distributors, etc. A mass-produced, scientifically-driven brand may still have romantic associations for the end consumer, for example.

Arguments agreeing with the question could have been that there now are fewer unknowns in wine since much has been explained by science (fermentation, etc), or that there is no romance in 'recipe' wines, high-volume brands, or controlled winemaking. Arguments against the question could have included the fact that most consumers still have no idea how wine is really made and love to retain the romantic associations they have. Many aspects of terroir remain unknown, and how wine ages is still largely a mystery. There will always be romance when a wine is associated with special birthdays or historical events. Perhaps the revival in interest in 'authenticity' has brought back the romantic notion of wine as being a pure product of the soil, and less controlled by science. Very few candidates considered a broad definition of romance and struggled to come up with sufficient material for a convincing argument.

The examiners commented that this is the type of question that demonstrates the importance of good essay writing. Many answers lacked structure. Some candidates used headings and bullet points, which rarely work in candidates' favour on this paper.

3. How relevant is blind tasting in assessing wine?

Key understandings:

- *The relevance of blind tasting to different groups: consumers, buyers, other wine professionals, producers, journalists, judges, etc.*
- *The absolute usefulness and limitations of blind tasting*

This was another straightforward question that nevertheless tripped up a good portion of candidates who neglected to discuss the relevance of blind tasting, and instead simply listed the pros and cons. Those who did answer the question, however, generally scored well, and this question had the highest pass rate of the paper.

We were looking for a thorough exploration of the practice, usefulness and limitations of blind tasting, and many candidates provided this, although surprisingly few used the example of their recent experience of the MW practical exam to illustrate their answer.

4. What are the causes for optimism in the wine world today?

Key understandings:

- *Ability to look at the multifaceted nature of the world of wine and identify current or upcoming elements within it that could lead to beneficial outcomes for consumer, producers and intermediaries.*
- *Evidence of widespread global knowledge of various aspects of the world of wine.*
- *Personal insight and ability to comment on future trends based on the current situation.*

This question gave candidates the opportunity to show personal insights and speculate on current trends, events and possibilities that might generate positive outcomes for any area of the wine world. Examples of causes for optimism could be: people drinking less, but better; booming wine education; a new generation valuing authenticity, thus encouraging greater sustainability in farming; greater awareness of the effects of agrochemicals, leading to cleaner wines and less polluted vineyards; the boom in craft beer with positive knock-on effects for the wine industry; new and untapped markets to exploit; positive research into wine's health benefits; the huge variety of wine available world-wide; the benefits of technology in creating reliable, high-quality product on a large scale; the possibility of growing vines in new places; the internet shifting power of information distribution to the end user, making for better informed consumers; new marketing possibilities in the digital age; wine preservation inventions increasing by-the-glass offerings; and many more.

There was plenty of scope for choice of material here and this was reflected in a good number of sound passes. The candidates who fell down usually did so because of simplistic arguments that were considered below MW standard, or factual errors. A number of candidates failed because they did not answer the question and gave reasons for pessimism instead of optimism. While it could have been a valid approach to the question to argue that there are zero causes for optimism in the wine world, this would have been a position that needed justification within the context of the question.

5. Does great wine need a great story?

Key understandings:

- *The role of stories in communicating about wine.*
- *What wine means to different people: from those who sell, make, buy, or drink it.*

This was not a popular question and the least well answered on the paper. Most candidates appeared to be unsure how to define 'great wine', so avoided doing so, which led to muddled answers that failed to reach a clear conclusion. The question was whether a great wine – however it is defined – needs a great story. In other words, is the story intrinsic to its greatness, or its perception as such? Questions that could have been addressed were: Can a great wine exist in a vacuum of information? Do those who sell, buy and drink it need more than simply the product? Do great wines make their own stories by virtue of their greatness? Is it necessary to know what a wine is (its history, provenance, etc) in order to appreciate it to the full? Would a great wine be as appreciated blind, or without its story? Are there any examples of great wines with no story? Or a wine made great by its story? Is the story only relevant for marketing, or is it intrinsic to the wine?

While there was scope for differing definitions of 'great' (from high-end fine wine to globally successful brands), there needed to be some definition and an underlying logic supported by examples. Some essays appeared to consider any wine to be 'great', which made it difficult to then come up with a convincing argument that answered the question. The examiners commented that essay style let many candidates down. The best answers showed clarity and focus, using examples to good effect.